

Computer Science Vocabulary

English for IT Professionals

Профессиональная лексика для программистов, аналитиков данных и IT-специалистов:
программирование, алгоритмы, базы данных, сети и кибербезопасность.

Двуязычный учебник / Bilingual Textbook

Английский текст · Объяснения на русском

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Предисловие

Данный учебник охватывает ключевую лексику финансового сектора: банковское дело, рынки капитала, управление рисками и корпоративные финансы.

По завершении курса вы сможете:

- Анализировать финансовые инструменты на английском языке
- Понимать деловые новости Bloomberg, Reuters, Financial Times
- Проходить собеседования в инвестиционных банках и финансовых компаниях
- Работать с финансовой документацией международных рынков

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MODULE 1 · COMPUTER SCIENCE & IT

The Financial System

Финансовая система

Learning Objectives

- 🔗 Describe the structure of the financial system
- 🔗 Explain the roles of central banks and commercial banks
- 🔗 Distinguish between financial markets and financial institutions
- 🔗 Use financial system vocabulary accurately

Introduction / Введение

The financial system is the network of institutions, markets, and instruments that channels funds from savers to borrowers and facilitates economic activity. A well-functioning financial system is essential for economic growth and stability.

RU:

Финансовая система — это сеть институтов, рынков и инструментов, которая перенаправляет средства от вкладчиков к заёмщикам. Хорошо функционирующая финансовая система необходима для экономического роста.

Theory / Теоретический материал

Structure of the Financial System

The financial system comprises three interconnected components:

1. FINANCIAL INSTITUTIONS

Banks, insurance companies, pension funds, and investment firms act as financial intermediaries — they accept deposits or premiums from savers and lend or invest those funds, earning a margin or fee.

Central Banks (e.g., the National Bank of Kazakhstan, the US Federal Reserve, the European Central Bank) stand at the apex of the banking system. They:

- Issue currency and control the money supply
- Set benchmark interest rates (the policy rate)
- Act as lender of last resort during banking crises
- Supervise commercial banks and maintain financial stability

Commercial banks take deposits, provide loans, process payments, and offer financial products.

2. FINANCIAL MARKETS

Markets where financial instruments are bought and sold:

- Money markets — short-term instruments (under 1 year): treasury bills, commercial paper
- Capital markets — long-term instruments (over 1 year): bonds, equities
- Foreign exchange (forex) markets — currency trading
- Derivatives markets — contracts based on underlying assets (futures, options, swaps)

3. FINANCIAL INSTRUMENTS

- Debt instruments: bonds, loans, mortgages
- Equity instruments: ordinary shares, preference shares
- Hybrid instruments: convertible bonds
- Derivatives: options, futures, forwards, swaps

The Interest Rate

The interest rate is the price of borrowing money. Central banks set a policy rate (e.g., the base rate, federal funds rate) which influences all other rates in the economy. When the central bank raises rates, borrowing becomes more expensive, cooling inflation but potentially slowing growth.

RU:

Финансовая система включает: (1) финансовые институты (банки, страховые компании); (2) финансовые рынки (денежные, фондовые, валютные); (3) финансовые инструменты (долговые, долевыe, деривативы). Центральный банк устанавливает базовую ставку — ключевую ставку процента, влияющую на всю экономику.

Professional Vocabulary / Профессиональная лексика

This module covers 48 professional terms.

Word / Term	Definition (EN)	Translation (RU)	Example
financial system (n.)	The network of institutions, markets, an...	финансовая система	A stable financial syst...
financial intermediary ...	An institution that channels funds from s...	финансовый посред...	Banks are the most co...
central bank (n.)	The institution responsible for monetary...	центральный банк	The National Bank of ...
commercial bank (n.)	A bank that accepts deposits and provid...	коммерческий банк	Halyk Bank is the larg...
monetary policy (n.)	Central bank actions to manage money s...	денежно-кредитная...	Tight monetary policy ...
interest rate (n.)	The cost of borrowing money, expressed...	процентная ставка	A rise in interest rates ...
base rate (n.)	The key policy rate set by a central bank,...	базовая (ключевая) с...	The NBK raised the ba...
inflation (n.)	The general increase in prices over time,...	инфляция	The central bank targ...
deflation (n.)	A general decrease in prices	дефляция	Deflation can trigger e...
money supply (n.)	The total amount of money circulating in...	денежная масса/пре...	Quantitative easing in...
liquidity (n.)	The ease with which assets can be conve...	ликвидность	Cash is the most liquid...
solvency (n.)	The ability to meet long-term financial o...	платёжеспособность	A bank must remain s...

Word / Term	Definition (EN)	Translation (RU)	Example
capital markets (n.pl.)	Markets for long-term financial instruments	рынки капитала	Companies raise funds on capital markets.
money markets (n.pl.)	Markets for short-term financial instruments	денежные рынки	The money market provides short-term financing.
stock exchange (n.)	An organised market where shares are traded	фондовая биржа	The AIFC (Astana International Financial Centre) is a stock exchange.
bond (n.)	A debt security issued by governments or corporations	облигация	Government bonds are sold to raise money.
equity (n.)	Shares in a company; ownership stake	акционерный капитал	Equity investors share in the profits.
dividend (n.)	A portion of company profits distributed to shareholders	дивиденд	The company paid a dividend to its shareholders.
yield (n.)	The income return on an investment as a percentage of the investment	доходность	The bond yields 7% per annum.
risk (n.)	The possibility of loss or underperformance	риск	Higher risk should be compensated by higher returns.
return (n.)	The gain or loss on an investment over a period of time	доходность/прибыль	The portfolio achieved a return of 15%.
portfolio (n.)	A collection of financial investments held by an individual or institution	инвестиционный портфель	Diversification in a portfolio reduces risk.
diversification (n.)	Spreading investments across different assets to reduce risk	диверсификация	Diversification is a core principle of portfolio management.
credit risk (n.)	The risk that a borrower will default on a loan	кредитный риск	Banks charge higher interest rates to cover credit risk.

Word / Term	Definition (EN)	Translation (RU)	Example
market risk (n.)	The risk of losses due to movements in ...	рыночный риск	Market risk includes i...
systemic risk (n.)	The risk that failure of one institution tri...	системный риск	The 2008 financial cris...
hedge (v./n.)	To reduce risk by taking an offsetting pos...	хеджировать; хедж...	Companies hedge aga...
derivative (n.)	A financial contract whose value depend...	производный финан...	Options and futures ar...
option (n.)	The right (but not obligation) to buy or s...	опцион	A call option gives the ...
futures contract (n.)	An agreement to buy or sell an asset at a...	фьючерсный конт...	Oil producers use futu...
swap (n.)	An agreement to exchange financial obli...	своп	An interest rate swap ...
leverage (n.)	Using borrowed money to increase the p...	финансовый рычаг/л...	High leverage amplifie...
gearing (n.)	The ratio of debt to equity in a company'...	коэффициент фина...	High gearing means t...
credit rating (n.)	An assessment of a borrower's creditwo...	кредитный рейтинг	Moody's, S&P, and Fit...
default (n./v.)	Failure to repay a debt as agreed	дефолт; допускать ...	A sovereign default ca...
quantitative easing (n.)	Central bank policy of buying assets to in...	количественное смя...	The Fed used quanti ta...

Word / Term	Definition (EN)	Translation (RU)	Example
fiscal policy (n.)	Government decisions on taxation and s...	фискальная политика	Expansionary fiscal po...
GDP (abbr.)	Gross Domestic Product — total value of...	ВВП — валовой внут...	Kazakhstan's GDP gre...
exchange rate (n.)	The price of one currency in terms of an...	обменный курс	A weaker tenge make...
forex (n.)	Foreign exchange market — where curre...	форекс / валютный ...	The forex market trad...
IPO (abbr.)	Initial Public Offering — first sale of a co...	IPO — первичное пуб...	The company raised \$...
M&A (abbr.)	Mergers and Acquisitions — combining ...	слияния и поглощен...	M&A activity increase...
valuation (n.)	The process of determining the economi...	оценка стоимости	DCF analysis is a com...
DCF (abbr.)	Discounted Cash Flow — a valuation met...	метод дисконтиров...	DCF valuation require...
underwriter (n.)	A financial institution that guarantees to ..	андеррайтер	Goldman Sachs acted ...
prospectus (n.)	A legal document providing details of a s...	проспект эмиссии	Investors should read ...
benchmark (n.)	A standard index used to evaluate invest...	эталон/индекс-орие...	The S&P 500 is a com...
asset management (n.)	Professional management of investment...	управление активами	BlackRock is the world...

Case Study: The 2008 Global Financial Crisis — Lessons for Finance Professionals

CASE STUDY

The 2008 global financial crisis began in the US mortgage market and quickly spread to the entire global financial system. Understanding it is essential for any finance professional.

Key causes:

1. Subprime mortgages: Banks issued loans to borrowers with poor credit histories, assuming house prices would always rise.
2. Securitisation: These risky mortgages were packaged into complex instruments called Mortgage-Backed Securities (MBS) and Collateralised Debt Obligations (CDOs), and sold globally.
3. Excessive leverage: Banks operated with very thin equity cushions.
4. Regulatory failure: Regulators failed to detect systemic risks building up.
5. Contagion: When Lehman Brothers collapsed in September 2008, panic spread worldwide.

The consequences: Global GDP fell; unemployment rose; governments spent trillions on bank bailouts. New regulations (Basel III, Dodd-Frank) were introduced to prevent a repeat.

Discussion Questions

1. What is securitisation, and why did it amplify the crisis?
2. Explain how excessive leverage made the crisis worse.
3. What is "systemic risk" and how did it manifest in 2008?
4. What regulatory changes were introduced after the crisis?
5. How does this crisis relate to the concepts of credit risk and market risk?

Exercises / Упражнения

Exercise 1: Match the financial term with its correct description.

Match the financial term with its correct description.

(Соотнесите термин с определением.)

- | | |
|---------------|---|
| 1. Bond | a) Right to buy/sell an asset at a set price |
| 2. Equity | b) Ownership stake in a company |
| 3. Option | c) Debt security with fixed interest payments |
| 4. Derivative | d) Contract based on an underlying asset |
| 5. Dividend | e) Profit distribution to shareholders |

Exercise 2: Complete with the correct word: inflation, deflation, yield, leverage, hedge.

Complete with the correct word: inflation, deflation, yield, leverage, hedge.

(Заполните пропуски правильными словами.)

1. When prices fall consistently, the economy experiences _____.
2. The bond's _____ fell as its price rose.
3. The company used currency options to _____ against forex risk.
4. High _____ can amplify both gains and losses.
5. Rising _____ erodes the purchasing power of money.

Professional Note

The CFA (Chartered Financial Analyst) qualification, administered by the CFA Institute, is the gold standard for finance professionals globally. All CFA materials are in English. Achieving CFA Level I requires mastery of financial statement analysis, economics, equity, fixed income, and derivatives vocabulary covered in this textbook.

Квалификация CFA (Chartered Financial Analyst) — золотой стандарт в мировых финансах. Все материалы CFA на английском языке. Сдача уровня CFA I требует уверенного владения лексикой финансового анализа, которая охвачена в этом учебнике.

Writing in Computer Science & IT

Professional writing in English follows specific conventions. Below are the most important genres and structures in your field.

Professional Emails (Деловые письма)

Emails in professional settings follow a clear structure: subject line, greeting, purpose, details, call to action, closing.

RU:

Деловое письмо на английском: тема, приветствие, цель, детали, призыв к действию, завершение.

EXAMPLE

Subject: Request for Meeting

Dear Mr Johnson,

I am writing to request a meeting to discuss the Q3 results.
Would you be available on Thursday at 2:00 PM?

Kind regards,
[Your name]

Professional Reports (Отчёты)

Reports use clear headings, factual language, and logical organisation: executive summary, background, findings, analysis, recommendations.

RU:

Отчёты: резюме для руководства, исходная информация, результаты, анализ, рекомендации.

EXAMPLE

REPORT: Q3 Financial Performance

Executive Summary
Revenue increased 8.3% year-on-year...

Key Findings

1. Revenue: +8.3% YoY
2. Operating profit: +5.7% YoY

Pronunciation Guide / Произношение

Correct pronunciation of Computer Science & IT terms is essential for professional credibility.

RU:

Правильное произношение терминов важно для профессионального имиджа. Ниже — типичные трудности.

1. Multi-syllable stress: ac-COUNT-ing, fi-NANCE / FI-nance, man-AGE-ment
2. Silent letters: debt /det/, receipt /rɪsi:t/, subtle /sʌtl/
3. British vs American: schedule /ʃedju:l/ (UK) vs /skedju:l/ (US)
4. Abbreviations: IFRS, GDP — pronounce each letter individually
5. Numbers: 8.5% = "eight point five percent" (not "comma")

Complete Professional Glossary / Профессиональный глоссарий

48 terms, alphabetically.

asset management — управление активами. Professional management of investments on behalf of clients

base rate — базовая (ключевая) ставка. The key policy rate set by a central bank, influencing all other rates

benchmark — эталон/индекс-ориентир. A standard index used to evaluate investment performance

bond — облигация. A debt security issued by governments or corporations to borrow money from investors

capital markets — рынки капитала. Markets for long-term financial instruments (bonds, equities)

central bank — центральный банк. The institution responsible for monetary policy and currency issuance

commercial bank — коммерческий банк. A bank that accepts deposits and provides loans to individuals and businesses

credit rating — кредитный рейтинг. An assessment of a borrower's creditworthiness (e.g., AAA, BBB, CCC)

credit risk — кредитный риск. The risk that a borrower will default on a debt obligation

DCF — метод дисконтированных денежных потоков. Discounted Cash Flow — a valuation method based on present value of future cash flows

default — дефолт; допускать дефолт. Failure to repay a debt as agreed

deflation — дефляция. A general decrease in prices

derivative — производный финансовый инструмент (дериватив). A financial contract whose value depends on an underlying asset

diversification — диверсификация. Spreading investments across different assets to reduce risk

dividend — дивиденд. A portion of company profits distributed to shareholders

equity — акционерный капитал/акции. Shares in a company; ownership stake

exchange rate — обменный курс. The price of one currency in terms of another

financial intermediary — финансовый посредник. An institution that channels funds from savers to borrowers

financial system — финансовая система. The network of institutions, markets, and instruments that channels funds through the economy

fiscal policy — фискальная политика. Government decisions on taxation and spending

forex — форекс / валютный рынок. Foreign exchange market — where currencies are bought and sold

futures contract — фьючерсный контракт. An agreement to buy or sell an asset at a future date and price

GDP — ВВП — валовой внутренний продукт. **Gross Domestic Product** — total value of goods and services produced in a country

gearing — коэффициент финансового рычага (брит.). The ratio of debt to equity in a company's capital structure (UK term)

hedge — хеджировать; хеджирование. To reduce risk by taking an offsetting position; a risk-reducing strategy

inflation — инфляция. The general increase in prices over time, reducing purchasing power

interest rate — процентная ставка. The cost of borrowing money, expressed as a percentage per year

IPO — IPO — первичное публичное размещение акций. **Initial Public Offering** — first sale of a company's shares to the public

leverage — финансовый рычаг/леверидж. Using borrowed money to increase the potential return on investment

liquidity — ликвидность. The ease with which assets can be converted to cash

M&A — слияния и поглощения (M&A). **Mergers and Acquisitions** — combining or taking over companies

market risk — рыночный риск. The risk of losses due to movements in market prices

monetary policy — денежно-кредитная политика. Central bank actions to manage money supply and interest rates

money markets — денежные рынки. Markets for short-term financial instruments (under 1 year)

money supply — денежная масса/предложение денег. The total amount of money circulating in an economy

option — опцион. The right (but not obligation) to buy or sell an asset at a set price

portfolio — инвестиционный портфель. A collection of financial investments held by an investor

prospectus — проспект эмиссии. A legal document providing details of a securities offering to potential investors

quantitative easing — количественное смягчение. Central bank policy of buying assets to inject money into the economy

return — доходность/прибыль от инвестиций. The gain or loss on an investment over a period

risk — риск. The possibility of loss or underperformance relative to expectations

solvency — платёжеспособность. The ability to meet long-term financial obligations

stock exchange — фондовая биржа. An organised market where shares are traded

swap — своп. An agreement to exchange financial obligations (e.g., interest payments)

systemic risk — системный риск. The risk that failure of one institution triggers collapse of the entire system

underwriter — андепайтер. A financial institution that guarantees to purchase unsold securities in

an offering

valuation — оценка стоимости. The process of determining the economic value of an asset or company

yield — доходность. The income return on an investment as a percentage of its price

Answer Key / Ответы

Module 1: The Financial System

Exercise 1

1. (see answer sheet)
2. (see answer sheet)
3. (see answer sheet)
4. (see answer sheet)
5. (see answer sheet)

Exercise 2

1. (see answer sheet)
2. (see answer sheet)
3. (see answer sheet)
4. (see answer sheet)
5. (see answer sheet)

Professional Resources

1. Financial Times ([ft.com](https://www.ft.com)) — authentic financial journalism for C1 level
2. Bloomberg ([bloomberg.com](https://www.bloomberg.com)) — real-time market data and analysis
3. CFA Institute ([cfainstitute.org](https://www.cfainstitute.org)) — free study materials
4. Investopedia Finance Academy — structured finance courses
5. KHAMADI ENGLISH Finance Flashcard Deck — 840+ terms